

Placement Portal Project Report (MAD 1- Project)

1. Student Details

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2. Project Details

Placement Portal – A Web Application for Managing Campus Recruitment

Problem Statement

Campus placements involve coordination between students, companies, and administrators. In many cases, managing job postings, student applications, and approvals manually can become time-consuming and difficult to track. There is a need for a centralized system that allows students to view job opportunities and apply for them, companies to post job openings, and administrators to monitor.

Approach to the Problem

To address this problem, a web-based placement portal was developed using the Flask framework with a modular structure using VS code. The system is designed with three different types of users: students, companies, and administrators.

Students can register on the portal, create their profiles, and view available job opportunities posted by companies. They can apply for jobs and track the status. Companies can create company profiles and post placement drives with job details. They can also view applications submitted by students. The administrator manages the overall system by approving student and company registrations and monitoring placement drives

3. AI / LLM Declaration

Used AI tool ChatGPT during the development of this project as a learning and assistance tool. This tool helped in understanding certain concepts, debugging some areas of code, and improving the structure and look of the web application.

However, the design decisions, coding, testing, and integration of the project were carried out manually and the final implementation was completed independently.

The extent of AI/LLM usage is around **15–20%**, limited to **code suggestions from VS code, UI styling and code debugging**.

4. Frameworks and Libraries Used

Technology / Library	Purpose
Flask	Core backend web framework
Flask-SQLAlchemy	Database integration for Flask
SQLAlchemy	ORM used to interact with the SQLite database
Jinja2	Template engine for rendering HTML pages
Bootstrap	Frontend styling and responsive UI
SQLite	Lightweight database for storing data
python-dotenv	Used to manage environment variables

5. Database ER Diagram

Tables: User, Student, Company, Placements, Applications

Relationships overview

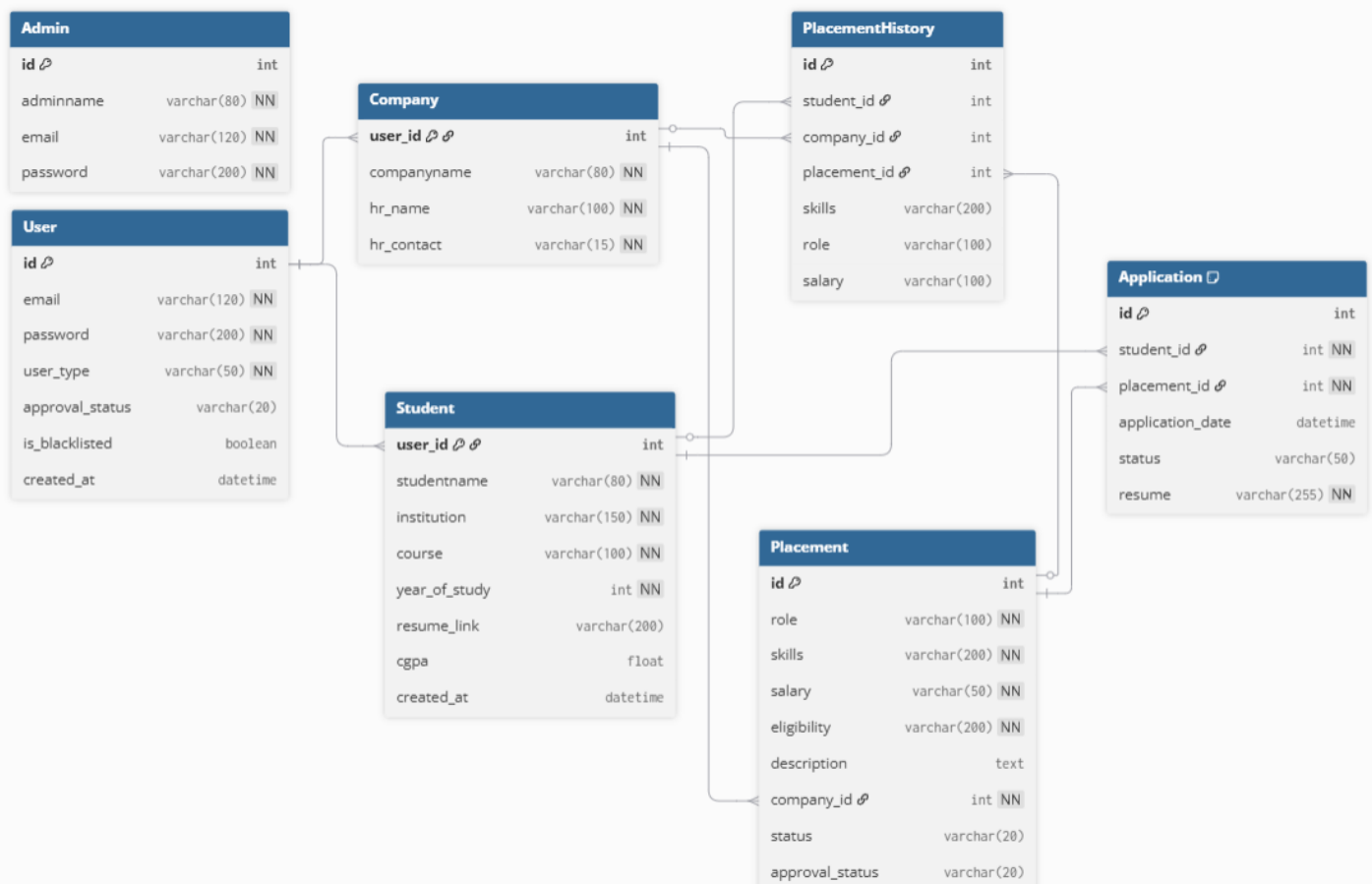
One-to-One → User → Student

One-to-One → User → Company

One-to-Many → Company → Placement

One-to-Many → Student → Application

One-to-Many → Placement → Application



6. API Resource Endpoints

The application contains several routes that handle different functionalities.

Authentication

- /login – Allows users to log in
- /register – Allows new users to register
- /logout – Logs the user out of the system

Student

- /student/dashboard – Displays student dashboard
- /student/jobs – Shows available jobs
- /student/apply/<id> – Allows student to apply for a job
- /student/profile – View or update student profile

Company

- /company/dashboard – Displays company dashboard
- /company/job/new – Post a new job opening
- /company/applications – View student applications
- /company/profile – View or update company profile

Admin

- /admin/dashboard – Admin overview page
- /admin/students – Manage student accounts
- /admin/companies – Manage company accounts
- /admin/placements – Manage placement drives
- /admin/applications – View all applications

7. Architecture Overview

- **app.py** – Main Flask application entry point handling server execution.
- **models.py** – Defines database tables and relationships using SQLAlchemy ORM.
- **routes.py** – Contains Flask routes for admin, student, company, placements, and applications.
- **templates/** – Jinja2 HTML templates used for dashboards, forms, and pages.
- **static/** – Stores CSS, JavaScript, and Bootstrap resources for UI styling.

Implemented Features

- User registration and login system.
- Role-based access for **Admin, Company, and Student**.
- Student profile creation and job application system.
- Company placement drive creation and management.
- Admin approval system for companies and placements.
- Application status tracking (**Applied, Shortlisted, Interview, Rejected, Placed**).
- Responsive user interface built with Bootstrap.

Additional Features

- Admin ability to **blacklist or activate users**.
- **Placement history tracking** for successfully placed students.
- Search functionality for students, companies, and placements.

7. Project Demonstration Video

Drive Link:

https://drive.google.com/file/d/1Pj4xJmiJmHehUg6a_2_elu0DEB5Swq5n/view?usp=sharing